

Nonlinear Hybrid Horizon Budgets

Exact Propagation of Recursive Weak-Mode Quotient Losses

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Abstract

RH-96 certifies the one-step energy loss from omitting weak projected-cross directions, but more aggressive locally valid cutoffs still fail at the endpoint. We derive an exact horizon decomposition that does not linearize the recursive Ritz map.

Let F_t denote the full-width update, A_t an adaptive update, x the source packet, and J the endpoint tail. Define the hybrid composition

$$H_j = F_N \circ \cdots \circ F_{j+1} \circ A_j \circ \cdots \circ A_1.$$

Our nonlinear hybrid telescoping theorem is the identity

$$J(H_N x) - J(H_0 x) = \sum_{j=1}^N [J(H_j x) - J(H_{j-1} x)].$$

It requires no linearity, differentiability, or common intermediate packet. The triangle inequality gives an absolute propagated horizon budget. If $A_j = F_j$ except at sparse omission times, only those hybrid endpoints need be replayed.

A 384-bit audit applies the identity to the three RH-96 cutoffs. All ten channel decompositions telescope and match the adaptive endpoint. For the primary 10^{-8} cutoff, the five omissions have worst absolute propagated budget only 1.0035×10^{-5} of the reference tail; the worst signed endpoint shift is 7.06×10^{-7} . The failing cutoffs are explained quantitatively: their worst absolute budgets are 0.024921 at 10^{-6} and 0.014092 at 10^{-4} , each crossing the one-percent endpoint allowance in exactly one channel. Some hybrid contributions are negative, so future full refresh can reverse a local quotient loss.

The endpoint threshold is therefore not mysterious: it is a propagated budget threshold. Hybrid replay is exact but a posteriori. The next task is a replay-free block propagation envelope. No uniform repeated-block theorem, Hilbert–Polya operator, zero identification, or Riemann Hypothesis result is claimed.

Keywords: nonlinear telescoping; hybrid composition; Duhamel principle; Ritz recursion; endpoint error budget; validated numerics.

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1 Introduction

The gap-weighted theorem of RH-96 answers a local question: how much captured energy can be lost when a weak complement direction is omitted [5]? At relative cutoff 10^{-8} , five such omissions are harmless over the complete source-to-endpoint chain. At cutoffs 10^{-6} and 10^{-4} , every local loss remains certified, yet one endpoint fails.

A natural response is to postulate a linear recurrence

$$e_t \leq \rho_t e_{t-1} + \varepsilon_t.$$

That form is useful eventually, but deriving a correct ρ_t for a recursive Ritz map is difficult. The enrichment subspace depends nonlinearly on the incoming packet through a projected cross SVD; different width choices produce different packets, which then produce different future subspaces.

This paper takes a more elementary route. Replace one adaptive decision at a time while replaying all later maps at full width. The endpoint values of these hybrid chains telescope exactly. This is the discrete nonlinear counterpart of a Duhamel or variation-of-constants decomposition [1, 2], but no derivative or tangent equation is needed.

The decomposition has three advantages.

1. It attributes an exact endpoint contribution to each adaptive decision, including its complete nonlinear future propagation.
2. It detects signed cancellation or reversal.
3. It turns the empirical endpoint frontier into a transparent absolute budget.

Its limitation is equally clear: it requires future replay. RH-97 therefore serves as an exact finite bridge between local quotient certificates and the next desired analytic object, a reusable block propagation envelope.

2 Nonlinear hybrid telescoping theorem

Let $X_0, \dots, X_N$ be arbitrary sets. For each time t , let

$$F_t, A_t : X_{t-1} \rightarrow X_t$$

be full and adaptive maps. Let $x \in X_0$ and let $J : X_N \rightarrow \mathbb{R}$ be any endpoint functional. No vector-space structure is required.

Definition 2.1 (Hybrid composition). For $j = 0, \dots, N$, define

$$H_j = F_N \circ \dots \circ F_{j+1} \circ A_j \circ \dots \circ A_1, \quad (1)$$

where $H_0 = F_N \circ \dots \circ F_1$ and $H_N = A_N \circ \dots \circ A_1$.

Theorem 2.2 (Nonlinear hybrid telescoping theorem). *For arbitrary maps and endpoint functional,*

$$J(H_N x) - J(H_0 x) = \sum_{j=1}^N \Delta_j, \quad \Delta_j := J(H_j x) - J(H_{j-1} x). \quad (2)$$

Proof. The right side is the telescoping sum of the scalar sequence $J(H_0 x), \dots, J(H_N x)$. No relation between the intermediate states of two different hybrids is needed. $\square$

The simplicity of the proof is the point. All nonlinear propagation is contained inside the endpoint evaluation of H_j .

Corollary 2.3 (Absolute propagated horizon budget).

$$|J(H_N x) - J(H_0 x)| \leq \sum_{j=1}^N |\Delta_j|. \quad (3)$$

Corollary 2.4 (Sparse-decision reduction). *Suppose $A_j = F_j$ outside an index set $S = \{j_1 < \dots < j_q\}$. Then $H_j = H_{j-1}$ for $j \notin S$, and*

$$J(H_N x) - J(H_0 x) = \sum_{\ell=1}^q [J(H_{j_\ell} x) - J(H_{j_\ell-1} x)]. \quad (4)$$

Thus only $q + 1$ endpoint values are required.

Remark 2.5. The hybrid contribution Δ_j is not necessarily positive, even if the adaptive one-step tail is larger than the full one-step tail. Later nonlinear full refreshes can contract, cancel, or reverse the injection.

3 Application to recursive Ritz refresh

For a fixed Gram sequence $G_1, \dots, G_N$, let F_t be the width-four projected-cross Ritz refresh. For a threshold τ , let A_t^τ choose the adaptive width of RH-96, between two and four. Both maps act on rank- r packet projectors. The endpoint functional is

$$J(V) = \mathcal{E}_{G_N}(V) = \text{tr } G_N - \text{tr}(V^* G_N V).$$

At one omission time j , RH-96 supplies a local tail injection

$$\varepsilon_j = \mathcal{E}_{G_j}(A_j^\tau V_{j-1}) - \mathcal{E}_{G_j}(F_j V_{j-1}) \geq 0.$$

RH-97 instead measures

$$\Delta_j = J(H_j V_0) - J(H_{j-1} V_0),$$

which includes all later full-width propagation. The ratio Δ_j/ε_j is a signed finite propagation multiplier. It is a diagnostic, not assumed a priori.

The absolute budget is normalized by the ambient leading-packet reference tail at time N :

$$B_\tau = \frac{\sum_{j \in S_\tau} |\Delta_j|}{\mathcal{E}_{G_N}(V_N^{\text{ref}})}. \quad (5)$$

The inherited endpoint allowance is $B_\tau < 0.01$.

4 Validated experiment

4.1 Protocol

We use the same ten channels, source seeds, rank clock, memory Gramians, and endpoints as RH-94–RH-96 [4]. For each threshold $10^{-8}, 10^{-6}, 10^{-4}$:

1. run the adaptive chain and record its omission times;
2. run the full-width endpoint H_0 ;
3. after each omission, take the actual adaptive prefix packet and replay all remaining updates at width four to obtain the next hybrid endpoint;
4. evaluate every endpoint tail as an exact-binary quadratic form in Arb at 384 bits [3].

The interval sum of hybrid contributions is checked against the direct adaptive-minus-full endpoint interval. The last hybrid is also checked against the adaptive endpoint.

4.2 Primary five-omission budget

All ten primary decompositions telescope, including the five channels with no omission, where the budget is exactly zero. The five nonzero contributions occur at the same updates isolated in RH-95 and RH-96.

The worst absolute horizon budget is

$$B_{10^{-8}} \leq 1.00343 \times 10^{-5}.$$

Table 1: Worst hybrid horizon budgets over ten channels.

cutoff	omissions	absolute budget	signed shift	green channels
10^{-8}	5	0.00001004	0.00000071	10
10^{-6}	11	0.02492070	0.02492070	9
10^{-4}	22	0.01409149	0.01409149	9

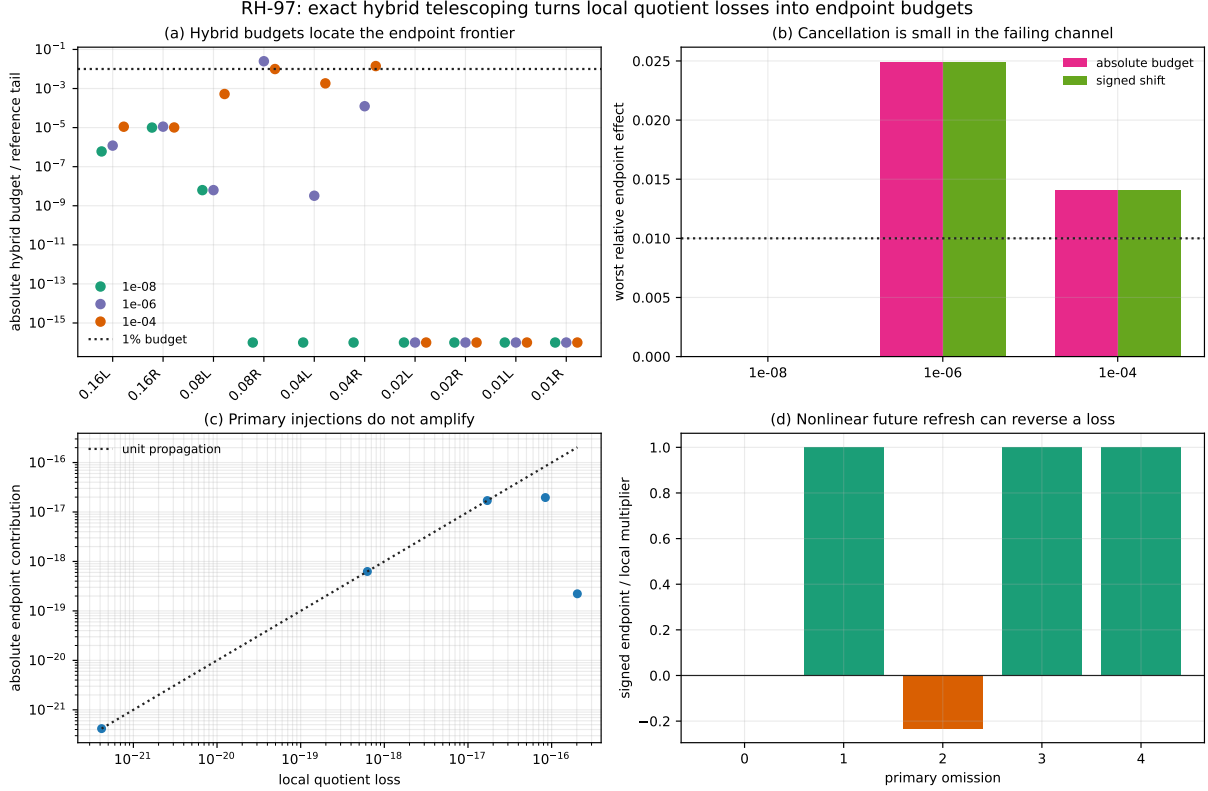


Figure 1: Exact nonlinear hybrid budgets. The primary cutoff stays far below one percent, while the two failed thresholds cross the inherited endpoint allowance in exactly one channel.

This is roughly three orders of magnitude below the one-percent allowance. The worst signed endpoint shift is smaller still:

$$\frac{|J(H_N x) - J(H_0 x)|}{\mathcal{E}_{G_N}(V_N^{\text{ref}})} \leq 7.06 \times 10^{-7}.$$

Two of the five propagated contributions are negative and three are positive. The largest absolute propagation multiplier is one; no primary local injection is amplified at the endpoint.

4.3 The failed thresholds

Table 1 explains the endpoint failures without appealing to a heuristic accumulation story. In the unique failing channel for each aggressive cutoff, the propagated absolute budget crosses one percent. The signed shift is nearly the absolute sum there, so cancellation does not rescue the endpoint.

Across all thresholds, some individual contributions are negative. The nonlinear future chain can improve the endpoint after a locally weaker choice. This is why a product of positive scalar local losses would be an incomplete description.

5 From hybrid replay to a block envelope

Hybrid replay solves the finite attribution problem exactly, but does not yet give an analytic all-level theorem. A reusable result would bound $|\Delta_j|$ from information available near the injection time. One target is

$$|\Delta_j| \leq \Gamma_{j,b} \varepsilon_j, \quad (6)$$

where $\Gamma_{j,b}$ is a propagation envelope for the future block b .

The audit suggests $\Gamma \leq 1$ for the tested omissions, but unit propagation is not a formal consequence of Ritz monotonicity: two chains enter future updates with different packets and different projected-cross spaces. RH-98 should therefore either:

1. derive a sufficient compressed spectral condition implying (6); or
2. construct a finite counterexample to unit propagation and identify the weakest corrected envelope.

A block envelope, combined with the local gap losses of RH-96, would produce a replay-free horizon budget and prepare the route for repeated blocks or a stopped exit clock.

6 Block grouping and verification structure

The scalar hybrid sequence can be grouped at any chosen block boundaries. Let

$$0 = n_0 < n_1 < \dots < n_B = N.$$

Define the block increment

$$\Delta^{(b)} = J(H_{n_b}x) - J(H_{n_{b-1}}x).$$

Then Theorem 2.2 immediately gives

$$J(H_Nx) - J(H_0x) = \sum_{b=1}^B \Delta^{(b)}, \quad |\Delta^{(b)}| \leq \sum_{j=n_{b-1}+1}^{n_b} |\Delta_j|. \quad (7)$$

This grouped identity is exact even when several adaptive decisions inside a block interact strongly. It provides the correct target for a future block envelope: one need not predict every individual signed contribution if a single compressed certificate controls $|\Delta^{(b)}|$.

There are two distinct ways to use blocks.

1. *Accounting blocks* merely aggregate already replayed hybrid increments through (7).
2. *Predictive blocks* would replace those replays by a bound derived from the block's initial packet, compressed spectra, and certified local injections.

RH-97 establishes the first exactly. RH-98 must supply the second or show why additional state variables are unavoidable.

6.1 Sparse replay cost

The naive hybrid family contains $N + 1$ chains. Sparse-decision reduction changes this to $q + 1$, where q is the number of actual omissions. In the primary audit $q = 5$ across all channels, rather than 120. At the two larger cutoffs, $q = 11$ and 22. A hybrid starting at omission time j replays only the suffix $j + 1, \dots, N$, so the work is proportional to

$$\sum_{j \in S} (N - j),$$

not qN in every channel. This remains a finite diagnostic cost, but it is not a continuum complexity theorem.

6.2 Interval closure

Each hybrid packet is selected by floating-point SVD, QR, and small symmetric eigensolves. Once selected, its endpoint quadratic form is interpreted as an exact binary input to Arb. For interval endpoint values $\mathbf{J}_j$, the audited contribution is the interval difference

$$\Delta_j = \mathbf{J}_j - \mathbf{J}_{j-1}.$$

The verification checks both

$$0 \in \sum_j \Delta_j - (\mathbf{J}_N - \mathbf{J}_0)$$

and equality of the final hybrid with the directly computed adaptive endpoint. The absolute budget uses the largest absolute endpoint of each Δ_j , so outward rounding cannot make a failing budget look smaller.

This protocol validates the endpoint values of the concrete selected packet chains. It does not validate the discontinuous SVD branch choice against all nearby real inputs; that stronger perturbation question belongs to the future block envelope.

7 Claim boundary

The nonlinear hybrid telescoping theorem and absolute propagated horizon budget are exact for arbitrary maps. The numerical conclusions concern the ten frozen packet chains. We do not prove an a priori Ritz-refresh Lipschitz law, a uniform block propagation envelope, repeated-horizon contraction, Stage-A closure, a Hilbert–Polya operator, zeta-zero identification, or the Riemann Hypothesis.

8 Conclusion

The local-to-global gap in adaptive Ritz enrichment can be closed exactly at finite horizon by hybrid compositions. Five primary omissions consume only 10^{-5} of the reference tail, while the failed cutoffs consume 1.4–2.5%. The endpoint frontier is therefore a propagated budget frontier.

The remaining issue is not how to attribute loss, but how to predict its future propagation without hybrid replay. A compressed block envelope is the next precise gate.

References

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